A Weighted Turán Unit-Step Inequality for Modified Bessel Functions and Its Determinant Corollary

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Abstract

For $x > 0$ and real $m \geq 1$ define, with I_μ the modified Bessel function of the first kind,

$$\varphi_m(x) \;=\; \frac{(m-1)\,I_{m-1}(x)^2 + (m+1)\,I_{m+1}(x)^2}{m\,I_m(x)^2}.$$

We prove that $\varphi_m(x) < \varphi_{m+1}(x)$ for every $x > 0$ and every real $m \geq 1$ — a weighted Turán-type monotonicity statement we have not found in the literature, although every ingredient of the proof is classical. The proof is fully elementary and uses exactly the arsenal of the companion note on unit-step order-monotonicity: the three-term recurrence and the Amos-type bound $\rho_m < x/(a+\sqrt{a^2+x^2})$, $a = m+\frac{1}{2}$, for $\rho_m = I_{m+1}/I_m$. The proof uses an exact factorization: with $u = \rho_m(x)$, $c = 1/x$, $S = u + 1/u$, $P = 1/u - u$,

$$\tfrac{1}{2}\big(\varphi_{m+1} - \varphi_m\big) \;=\; (S - 3c)\,\big(P - (2m+1)c\big) \;+\; (2m+1)\,c^2,$$

an algebraic identity once ρ_{m-1} and ρ_{m+1} are eliminated by the recurrence. The second factor is positive by the calibrated Amos bound — it is precisely the unit-step inequality of the companion note — and the first because the same bound forces $u < x/(2m+1) \leq x/3$. As an application we obtain the strict ordering $c_{mn} < 0$ $(m < n)$ of the 2×2 determinants of the coefficient arrays of a π-local surface expansion arising in an audit-first programme on two-dimensional $SU(2)$ lattice gauge theory; this is the structural half of a positivity problem whose remaining half — a signed-series positivity question — we state precisely and leave open. No numerical cancellation rate is asserted in this revision. An accompanying earlier Lean formalization encodes the ordered-field implication with the analytic recurrences and Amos bound as named hypotheses. Its archived verification record is supplied separately; no new Lean or numerical run is claimed by this revision.

1 Statement

Throughout, $x > 0$, I_μ is the modified Bessel function of the first kind, all values $I_\mu(x)$ for $\mu \geq 0$ are positive, and $\rho_\mu(x) = I_{\mu+1}(x)/I_\mu(x) \in (0,1)$.

Theorem 1 (weighted Turán-type unit-step monotonicity)**.** *For every $x > 0$ and every real $m \geq 1$,*

$$\varphi_m(x) \;=\; \frac{(m-1)\,I_{m-1}(x)^2 + (m+1)\,I_{m+1}(x)^2}{m\,I_m(x)^2} \;<\; \varphi_{m+1}(x), \tag{1}$$

strictly. Equivalently, in terms of the consecutive ratios $\rho_\mu = I_{\mu+1}/I_\mu$:

$$\frac{1}{m}\Big(\frac{m-1}{\rho_{m-1}^2} + (m+1)\rho_m^2\Big) \;<\; \frac{1}{m+1}\Big(\frac{m}{\rho_m^2} + (m+2)\rho_{m+1}^2\Big).$$

The weights $(m-1, m, m+1)$ are not decorative: they arise from a specific surface expansion (Section 3), and — as the proof shows — this choice admits the exact factorization below over the classical Amos bound. No uniqueness classification of weights is asserted here; compare [1, 2, 3]. We have not found (1) stated in the Turán-type literature for modified Bessel functions [4, 5, 6], though it sits squarely in its tradition: for fixed x, log-concavity of $\mu \mapsto I_\mu(x)$ (the classical Turán inequality [4, 5]) controls the *first-order* behaviour of the ratio sequence, while (1) is a statement about a *weighted second-order* combination.

2 The proof

We use two classical facts, valid for all real orders ([7, 10.29.1]): the three-term recurrence, in the two forms

$$\frac{1}{\rho_{m-1}} = \rho_m + \frac{2m}{x}, \qquad \rho_{m+1} = \frac{1}{\rho_m} - \frac{2(m+1)}{x}, \tag{2}$$

and the classical Amos-type upper bound [1, 2]. Theorem 2 of Ruiz-Antolín and Segura [3], applied with their order $\nu = m+1 \geq 2$, supplies exactly the bound used here:

$$\rho_m(x) \;<\; U_m(x) := \frac{x}{a+s}, \qquad a = m + \tfrac{1}{2}, \quad s = \sqrt{a^2 + x^2}. \tag{3}$$

Step 1: elimination

Write $u = \rho_m(x) \in (0,1)$ and $c = 1/x$. Substituting (2) into the ρ-form of (1) and clearing the positive denominators $m, m+1$, the claim $\varphi_{m+1} - \varphi_m > 0$ becomes $\Delta_m > 0$ where

$$\Delta_m \;=\; \frac{m^2}{u^2} - (m^2-1)\Big(u + 2mc\Big)^2 + m(m+2)\Big(\frac{1}{u} - 2(m+1)c\Big)^2 - (m+1)^2u^2.$$

Step 2: the exact factorization

Lemma 2. *With $S = u + 1/u$ and $P = 1/u - u$,*

$$\tfrac{1}{2}(\varphi_{m+1} - \varphi_m) \;=\; \frac{\Delta_m}{2m(m+1)} \;=\; (S - 3c)\,(P - (2m+1)c) \;+\; (2m+1)\,c^2.$$

Proof. Expanding Δ_m and collecting, the coefficients of $1/u^2$ and u^2 are $\pm 2m(m+1)$, the linear-in-c terms are $-4m(m+1)c\,[(m-1)u + (m+2)/u]$, and the c^2 terms sum to $8m(m+1)(2m+1)c^2$; dividing by $2m(m+1)$,

$$\frac{\Delta_m}{2m(m+1)} = \Big(\frac{1}{u^2} - u^2\Big) - \frac{2}{x}\Big((m-1)u + \frac{m+2}{u}\Big) + \frac{4(2m+1)}{x^2}.$$

Now $(m-1)u + (m+2)/u = \frac{1}{2}[(2m+1)S + 3P]$ and $1/u^2 - u^2 = PS$, so the right side equals $PS - c(2m+1)S - 3cP + 4(2m+1)c^2 = (S-3c)(P-(2m+1)c) + (2m+1)c^2$. □

Step 3: positivity of both terms

The second factor is strictly positive: $P-(2m+1)c>0$ is *exactly* the unit-step order-monotonicity inequality $1/\rho_m-\rho_m>(2m+1)/x$ of the companion note [8], which follows from (3) by the exact calibration $1/U_m-U_m=(2a)/x=(2m+1)/x$ together with the strict decrease of $t\mapsto 1/t-t$ on $(0,\infty)$.

The first factor is strictly positive: since $s>a$ in (3), $U_m<x/(2a)=x/(2m+1)$, so $u<x/(2m+1)\le x/3$ for $m\ge 1$, hence $S>1/u>(2m+1)/x\ge 3/x=3c$.

The third term $(2m+1)c^2$ is positive. This proves Theorem 1. □

Remark 3 (the calibration, for the third time). The bound U_m of (3) is the positive root of $1/t-t=(2m+1)/x$: it is the extremal function of the threshold that appears in the factorization. In the companion note [8] this exact calibration was the engine of the unit-step monotonicity of $(\log I_\nu)'$ and was shown to be *equivalent* to the Amos bound; here the same calibrated threshold reappears, multiplied against a second factor whose positivity is again a consequence of the same bound. Within the audit programme from which these notes emerge, this is the third lock opened by the same key, which suggests the calibration identity, rather than any particular inequality, is the structural invariant of this family of problems.

3 Application: ordering of the π-local surface expansion

The following objects arise in an audit-first research programme on the two-dimensional $SU(2)$ lattice gauge theory with Wilson action (Griffiths-II positivity on 2-complexes; see the programme records [9] for context). For $\beta>0$ let $r_k=I_{k+1}(\beta)/I_1(\beta)$, $k\ge 0$, and define the coefficient arrays ($k\ge 1$)

$$A_k=r_{k-1}^2\sum_{j\in\{k-2,\,k\},\ j\ge 0}(j+1)\,r_j^2,\qquad B_k=k\,r_{k-1}^4.$$

A direct computation shows $A_k/B_k=\varphi_k(\beta)$ for every $k\ge 1$ — including $k=1$, where the boundary truncation of the sum (the $j=-1$ term is absent) is annihilated by the weight $(k-1)$ in φ_k, so no separate edge case is needed.

Corollary 4 (determinant ordering)**.** *For all $\beta>0$ and all integers $1\le m<n$,*

$$c_{mn}\;=\;A_mB_n-A_nB_m\;=\;B_mB_n\,(\varphi_m(\beta)-\varphi_n(\beta))\;<\;0.$$

Proof. $B_k>0$, and $\varphi_m<\varphi_n$ by Theorem 1 iterated over unit steps from m to n. □

Lemma 5 (absolute convergence of the signed application series)**.** *For fixed $\beta>0$, the series defining $W(t)$ below is absolutely convergent, uniformly for real t.*

Proof. The positive power series for integer-order I_j [7, 10.25.2] gives, for $j\ge 0$,

$$I_j(\beta)=\sum_{h\ge 0}\frac{(\beta/2)^{j+2h}}{h!(j+h)!}\le\frac{(\beta/2)^j}{j!}\exp(\beta^2/4).$$

Thus r_k has a factorial tail for fixed β. From their definitions, A_k and B_k have finite polynomially weighted sums of every fixed order; the finitely many small indices, including $k=1$, cause no problem. Moreover $|c_{mn}|\le A_mB_n+A_nB_m$, while the absolute value of the trigonometric bracket

defining W is at most $2(m+n)$. The corresponding double sum is bounded by products of the finite zeroth and first moments of A_k and B_k. This gives a summable bound independent of t. Absolute convergence justifies the parity rewriting below; it supplies no sign for the resulting sum. □

Open problem 6 (what remains open, stated honestly). The positivity lemma that Corollary 4 serves is: for $t = \pi - \varepsilon$, $\varepsilon \in (0, 0.4]$, and all $\beta > 0$,

$$W(t) \;=\; \sum_{1 \le m < n} c_{mn} \left[(m-n)\sin((m+n)t) + (m+n)\sin((n-m)t) \right] \;<\; 0.$$

Writing $p = n - m$, $q = n + m$, the substitution $t = \pi - \varepsilon$ gives the *exact identity*, valid for every pair,

$$W = \sum_{m<n} |c_{mn}|\,(-1)^{m+n} \left[q\sin(p\varepsilon) - p\sin(q\varepsilon) \right],$$

where Corollary 4 supplied $c_{mn} = -|c_{mn}|$. For the pairs with $q\varepsilon \le \pi$ the bracket is strictly positive (decrease of $\sin(y)/y$ on $(0, \pi]$); for $q\varepsilon > \pi$ it is not sign-definite, so the sum is a signed-weight competition. The claim $W < 0$ is an *alternating-parity dominance* problem, exact in the rewriting and open in the dominance. Lemma 5 makes the displayed series well defined for each fixed β; it does not establish its sign. Earlier exploratory totals and precision prescriptions are not presented here as validated numerical data: their parameter, truncation and error specifications were incomplete. Cancellation may make naive summation unreliable, but it excludes neither a sufficiently sharp estimate-based proof nor alternative numerical methods. No universal decimal-precision threshold is claimed. A future numerical study must specify (β, ε), cutoffs, working precision, tail bounds and error control. The problem remains open. The coefficient arrays are the actual proved application; their role in a gauge-theory programme is context, not a derivation of a physical observable or a positivity theorem.

4 Machine verification

The accompanying historical record reports a Lean 4/Mathlib check of this ordered-field argument in `PhiMonotone.lean`. The recorded command is `lake env lean`, with Lean `4.30.0-rc2` and Mathlib revision

`cd3b69baae9cd81a572a3720f2372655eca39038`.

The recorded declarations are:

- `amos_calibration'`: the exact calibration;
- `amos_small`: the inequality $u < x/(2m+1)$ from (3);
- `phi_unit_step`: Lemma 2 and positivity;
- `phi_step_of_recurrences`: Theorem 1, with (2) and (3) as the named hypotheses `hrec1`, `hrec2`, `hAmos`.

The archived log, dated 9 July 2026, reports the following standard-axiom output; this revision has not replayed it:

```
'amos_calibration'' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'amos_small' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'phi_unit_step' depends on axioms:
    [propext, Classical.choice, Quot.sound]
'phi_step_of_recurrences' depends on axioms:
    [propext, Classical.choice, Quot.sound]
```

These are the archived run's reported exit code 0 and absence of `sorry` or `admit`, not new run results. As in the companion note, the encoded statement is the ordered-field implication [recurrences + Amos bound ⇒ monotonicity]; the analytic inputs themselves are classical and cited.

5 Reproducibility and computational scope

The supplied files are:

- `PhiMonotone.lean`;
- the unchanged historical log `LEAN-VERIFICATION-LOG.txt`.

Their programme source commit is

`871bb2bee8410dfb5b225d7961203eefaca5c15e`.

The source paths at that commit are

```
papers/phi-lemma/PhiMonotone.lean
papers/phi-lemma/LEAN-VERIFICATION-LOG.txt.
```

The historical log records the command

```
lake env lean PhiMonotone.lean
```

with Lean `4.30.0-rc2` and the full Mathlib revision displayed above. These are historical provenance, not a newly authenticated replay in that environment. The package `SOURCE207.json` binds the exact manuscript and supplied files by Git-LF hashes; the build record identifies the frozen source commit, actual Linux tool versions, process exits and exact PDF hash.

This revision makes no fresh symbolic or numerical verification claim. Earlier finite-grid experiments do not prove the real-order theorem, validate the open sign claim, or establish a universal working precision. The analytic argument and Lemma 5 are the evidence provided here. A formal rerun, if performed, must use a clean pinned Mathlib environment and retain complete exit and axiom logs. The recorded conditional theorem is not a formal construction of analytic Bessel functions or a complete gauge-theory argument.

6 Model assessments and review context

Codex assisted manuscript preparation and review orchestration. An AI assessment is not human peer review, formal verification, a priority ruling or editorial acceptance. Exact served

backend, reasoning effort and reviewer independence are not inferred from the product label. The current PDF is frozen before any new exact-artifact numerical review is issued; such a report is a separate hash-bound record, not a number inherited from an earlier version. Full issued reports, objections and actual review context must accompany any publication. The earlier v1 artifact and its review remain preserved; the present scientific changes address attribution, convergence, computational scope and reproducibility. No rating of v1 is claimed for this revision.

Acknowledgements

This note is part of an audit-first research programme whose records, Lean sources and numerical suites are public at https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME. The statement is presented as a lemma with an application and an open problem, in that order; nothing here is claimed about the open problem beyond its precise reduction, and nothing is claimed about four-dimensional gauge theory.

References

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[8] L. Eriksson, *An elementary recurrence–Amos proof of the unit-step order-monotonicity of* $(\log I_\nu)'$*, with a Feynman–Hellmann application to two-dimensional lattice gauge theory*, July 2026; https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME/tree/main/papers/bessel-amos-fh.

[9] L. Eriksson, *THE MASTER MAP: an audit-first navigation guide to the conditional construction of 4D SU(N) Yang–Mills with mass gap*, ai.viXra:2602.0096, and the programme repository https://github.com/lluiseriksson/THE-ERIKSSON-PROGRAMME.